AF-3957 **MMI1C1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

First Semester

Microbiology

GENERAL MICROBIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Mycoplasma.

2. Joseph Lister.

3. Mordant.

4. Numerical aperture.

5. Chemoautotrophs.

6. Disinfection.

7. O-antigen.

8. Prions.

9. MRSA.

10. *Desulfovibrio*.

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) What is the significance of Robert Koch's contributions to microbiology ?

(*Or*)

(b) Describe the salient features of archaebacteria.

12. (a) Explain the principle behind the working of the Phase Contrast microscopes.

(*Or*)

(b) Explain the various differential staining reactions.

13. (a) Explain in detail on the differential media.

(*Or*)

(b) Give a brief note on factors affecting bacterial growth.

14. (a) Explain in detail the Gram Positive bacterial cell structure.

(*Or*)

(b) Write a note on plasma membrane of gram negative bacteria.

15. (a) Describe the mode of drug resistance in microbes.

(*Or*)

(b) Write a note on mode of action of beta-lactum antibiotics.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Describe in detail the classification system proposed by Carl Woese.

17. Write an essay on electron microscopy. Describe the types and their working principles.

18. Describe in detail about the various methods of microbial preservation.

19. Describe in detail on the locomotory apparatus in bacteria and its functioning.

20. What are microbial toxins ? Classify them and explain their mode of action with examples.

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AF-3958 **MMI1C2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

First Semester

Microbiology

BIOCHEMISTRY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. Define Protein.

2. Write any two properties of Water.

3. Define Electron transport chain.

4. Define Simple diffusion.

5. Define Free energy.

6. Define O/R Potential.

7. What is Anabolism ?

8. Define Apoenzyme.

9. Define β oxidation.

10. Define Transmethylation.

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write an account on properties of Aminoacids.

(*Or*)

(b) Write notes on Hydrogen ion concentration.

12. (a) Explain the structure of Plasmamembrane.

(*Or*)

(b) Justify the first law of Thermodynamics.

13. (a) Briefly explain the microsomal electron transport chain.

(*Or*)

(b) Describe about Free energy.

14. (a) Briefly describe the biosynthesis of fattyacid.

(*Or*)

(b) Explain the gluconeogenesis.

15. (a) Write short notes on Riboenzyme.

(*Or*)

(b) Briefly explain the enzyme regulation.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Write an essay on biological functions of proteins.

17. Describe the mechanism of membrane transport.

18. Give an elaborate account on role of ATP in metabolism.

19. Explain the TCA Cycle.

20. Define Immobilization. Describe the methods of enzyme immobilization and add note on its application.

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AF-3959 **MMI1C3**

M.Sc. DEGREE EXAMINATION, APRIL 2011

First Semester

Microbiology

MICROBIAL PHYSIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define Enthalpy.

2. Define Bioenergetics.

3. Chlorophyll.

4. Rhodopsin.

5. Pasteur effect.

6. Substrate level phosphorylation.

7. Endospores.

8. Akinites.

9. Archaebacteria.

10. Quorum sensing.

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write a short note on electron carriers.

(*Or*)

(b) Give an account on artificial electron donors.

12. (a) What are the types and functions of bacteriochlorophylls ?

(*Or*)

(b) Describe the process of oxygenic photosynthesis.

13. (a) Describe the Embden–Meyerhoff–Paranas Pathway in detail.

(*Or*)

(b) Elaborate homolactic fermentation.

14. (a) Explain in detail the process of endospore formation.

(*Or*)

(b) Describe the formation of heterocysts.

15. (a) Give a brief account on extremophiles.

(*Or*)

(b) Write about the adaptive mechanisms in osmophilic organisms.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Write an essay on bioenergetics.

17. Explain in detail the Calvin cycle.

18. Define Fermentation. Explain the process with the possible pathways involved.

19. Give a detailed account on microbial differentiation.

20. Explain in detail about the mechanism of Quorum sensing and give their possible applications.

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AF-3960 **MMI1E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Microbiology

Elective I—COMPUTER APPLICATIONS AND BIOINFORMATICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Microprocessor.

2. Compilers.

3. RDBMS.

4. VoIP.

5. Gap penalty.

6. Substitution matrix.

7. ExPASy.

8. MP tree.

9. *Ab initio* approach.

10. RefSeq.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Describe about fifth generation computers.

(*Or*)

(b) Discuss on concept programming.

12. (a) Write short notes on modern networking.

(*Or*)

(b) Explain on basic concepts of indexing and hashing.

13. (a) List out some of the metabolism pathway databases.

(*Or*)

(b) Give an account on similarity searching.

14. (a) How will you predict tertiary structure of proteins ?

(*Or*)

(b) Differentiate between Cladogram and Dendrogram.

15. (a) Discuss on comparative genomics approach for gene prediction.

(*Or*)

(b) Briefly describe about protein-protein interactions.

Part C (3 × 10 = 30)

Answer any **three** of the following questions.

16. Elaborate on computer architecture and advances available today.

17. Discuss on the network security methods.

18. Write an essay on the methods of sequence alignment.

19. How will you determine the secondary and tertiary structure of proteins ? Illustrate with suitable examples.

20. Describe about DNA microarray technique and its applications.

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AF-3961 **MMI2C1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Second Semester

Microbiology

MOLECULAR BIOLOGY AND MICROBIAL GENETICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Auxotrophy.

2. Tetranucleotide hypothesis.

3. Azathioprine.

4. Photolyase.

5. Transformation efficiency.

6. Retrotransposons.

7. DnaB and DnaG.

8. Tus protein.

9. Structural genes.

10. TATA box.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Illustrate on Hershey and Chase experiments.

(*Or*)

(b) Discuss on the mutant nomenclature used in genetics.

12. (a) Write short notes on SOS response.

(*Or*)

(b) Explain on the mechanism of Ames test.

13. (a) How transposons are evolved ? Discuss.

(*Or*)

(b) Describe on the method of producing competence in bacteria

14. (a) Add a note on the method of inducing lysogeny in bacteriophage λ.

(*Or*)

(b) Elaborate on regulation of DNA replication.

15. (a) Briefly discuss on prokaryotic mRNA degradation

(*Or*)

(b) Explain on stringent response in bacteria.

Part C (3 × 10 = 30)

Answer any **three** of the following questions.

16. How DNA as the genetic material was proved ? Explain.

17. Elaborate on various types of DNA damage.

18. Write an essay on conjugation mechanisms.

19. Explain on the genome structure of bacteriophage lambda.

20. Discuss in detail about the mechanism of transcription and translation regulations in bacteria.

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AF-3962 **MMI2C2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Second Semester

Microbiology

ENVIRONMENTAL AND AGRICULTURAL MICROBIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Thermal inversion.

2. Green house effect.

3. Define Anaerobic fermentation.

4. AMS.

5. Composting.

6. Rhizoextraction.

7. Nitrifying bacteria.

8. Antagonism.

9. RTBV and RTSV.

10. Baculo Virus.

Part B

(5 × 5 = 25)

Answer **all** the questions.

11. (a) Write short notes on :

(i) Thermal inversion.

(ii) Photochemical smog.

(*Or*)

(b) Narrate the effects of air pollution on human health.

12. (a) Explain the primary treatment methods of industrial effluents.

(*Or*)

(b) Describe the biological effects of eutrophication process.

13. (a) Brief an account on mycoremediation.

(*Or*)

(b) What are the microbes involved in the degradation of chlorinated hydrocarbons ?

14. (a) Define Biogeochemical cycle. Elaborate the iron and manganese cycles.

(*Or*)

(b) Explain the nitrogen fixation process of free living microorganisms.

15. (a) Discuss in detail about any three bacterial disease of plants.

(*Or*)

(b) Summarize the "recent advances in biological pest control".

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Relate the air pollution and acid rain. Give the effects of acid rain on the ecology.

17. Give a descriptive process of domestic sewage treatment and waste disposal.

18. Discuss the organisms and mechanisms involved in pesticide degradation.

19. Write an essay on 'biological nitrogen fixation by symbiotic bacteria'.

20. Describe in detail about the symptoms, pathogenesis and control measures of fungal diseases of plants.

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AF-3963 MMI2C3

M.Sc. DEGREE EXAMINATION, APRIL 2011

Second Semester

Microbiology

FERMENTATION TECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Mutation.

2. What is primary screening ?

3. How do you immobilize the enzymes by covalent bonding ?

4. Define Fermentation.

5. Define Impeller.

6. What are the materials used for the construction of a fermentor ?

7. What are the composition of biogas ?

8. List out the uses of amylase.

9. Define Precipitant.

10. Define Centrifugation.

Part B

$(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Write short notes on industrially important microbes.

(*Or*)

(b) Briefly explain the basic principles of fermentation technology.

12. (a) Give an account on batch fermentation.

(*Or*)

(b) Write short notes on contour plot.

13. (a) Write short notes on Newtonian behaviour of fluid.

(*Or*)

(b) Briefly explain the membrane bioreactor.

14. (a) Briefly explain the citric acid production.

(*Or*)

(b) Write the procedure for beer production.

15. (a) Give an account on spray drier.

(*Or*)

(b) Briefly explain the foam separation.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe the protoplast fusion. Add a note on its significance.

17. Define Immobilization. Add a note on the immobilization of enzymes.

18. Write in detail about the design of a reactor.

19. Write an essay on single cell protein.

20. Give an account on fermentation economics and cost analysis.

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AF-3964 **MMI2E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Microbiology

Elective : MARINE MICROBIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Prions.

2. MTCC.

3. Barophiles.

4. RFLP.

5. *Ex situ* bioremediation.

6. WSSV.

7. Bacterio-rhodopsin.

8. Triose phosphate Isomerase.

9. Gene annotation.

10. Barcoding.

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write a short note on preservation of marine microbes.

(*Or*)

(b) Give an account on various culture collection centres for marine microbes.

12. (a) Elaborate on the adaptation of extremophiles.

(*Or*)

(b) Describe the process of oxygenic photosynthesis.

13. (a) Describe how microbes can be used in the biodegradation of pesticides.

(*Or*)

(b) Describe the role of marine microbes in the carbon cycle of the oceans.

14. (a) Describe the Entner Duodoroff Pathway in detail.

(*Or*)

(b) What are the types of functions of bacterio-chlorophylls ?

15. (a) Explain the process of bar coding marine microbes.

(*Or*)

(b) Write a brief note on the various types of mutations.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Write an essay on the modern techniques used in the classification of microbes.

17. Write an essay on the characteristics of extremophiles, their habitats and their importance.

18. Explain in detail any two of diseases caused by marine microbes.

19. Discuss in detail the Calvin cycle.

20. Explain in detail about the various DNA sequencing methods.

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AF-3965 **MMI3C1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Microbiology

MEDICAL BACTERIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. How do you culture the Peptostreptococcus ?

2. What are the uses of blood tellurite medium ?

3. What is Quelling reaction ?

4. Comment on Griffith typing.

5. Define Mantoux test.

6. What is Malta fever ?

7. Define Weils's disease.

8. What are the characteristics of Chlamydiae ?

9. Write the growth requirement of *Mycoplasma* and *Ureaplasma*

10. List the serodiagnostic methods for syphilis.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) What is Virulence ? Write a note on Virulence factors of bacterial pathogens.

(*Or*)

(b) Give an account on collection microscopic and bacteriological examination of sputum specimen.

12. (a) Describe the pathogenesis and laboratory diagnosis of tetanus.

(*Or*)

(b) Give an account on virulence factors and diseases causes by *Staphylococcus aureus*.

13. (a) Write a short note on diagnosis and treatment of Enteric fever.

(*Or*)

(b) Give an account on pathogenesis and diagnosis of bacillary dysentery

14. (a) What is Q fever ? How do you diagnose and treat the Q fever ?

(*Or*)

(b) How does cold agglutination test used in diagnosis of atypical pneumonia ?

15. (a) Differentiate the Gonococcus from Meningococcus.

(*Or*)

(b) Write a short note on serodiagnosis of STD.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. What is normal flora ? Schematically explain the normal flora types and functions in various parts of the Human body.

17. What are the diseases caused by *Staphylococcus aureus* ? Describe in detail on cultural and biochemical identification of *Staphylococcus aureus*.

18. What is Cholera ? Briefly describe pathogenesis and lab diagnosis of *Vibrio* infection.

19. Discuss in detail about the pathogenicity, transmission and diagnosis of Leptospirosis.

20. Explain the laboratory diagnosis of pyrexia of unknown origin.

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AF-3966 **MMI3C2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Microbiology

VIROLOGY, MYCOLOGY AND PARASITOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. List the antiviral agents.

2. What are viriods ?

3. What are the viral disease transmitted by mosquito ?

4. How do you diagnose the Rotavirus infection ?

5. Name the plant viruses transmitted by seeds.

6. What is potato leaf roll ?

7. How do you cultivate the *Entamoeba histolytica* ?

8. Differentiate the *Trypanosoma brucei* from *Trypanosoms cruzi*.

9. Define Oral thrush.

10. What is otomycosis ?

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write a note on cultivation methods of viruses.

(*Or*)

(b) Summarize the serodiagnostic methods of viral infection.

12. (a) Describe the pathogenesis and diagnosis of Influenza virus.

(*Or*)

(b) Give an account on properties and diseases caused by oncogenic viruses.

13. (a) Write a short note on prevention methods of viral disease in crop plants.

(*Or*)

(b) Give an account on pathogenesis, transmission, symptoms and control of TMV.

14. (a) Give an account on Hook work infection.

(*Or*)

(b) Briefly write a note on *Leishmania donovani* infection.

15. (a) Give an account on the Mycetoma infections.

(*Or*)

(b) Write a note on pathogenesis and diagnosis of crytococcosis.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about the viral vaccine types with examples.

17. Describe the pathogenesis, transmission and lab diagnosis of HIV infection.

18. Briefly describe different methods of plant virus transmission.

19. Discuss in detail about the pathogenesis, transmission, diagnosis of *Wucheriria bancrofti* infection.

20. What is dermatophytosis ? Explain the clinical types of dermatophytosis.

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AF-3967 **MMI3C3**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Microbiology

IMMUNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Who discovered the human ABO blood group system?

2. What is convalescent serum ?

3. Comment of HAT selection.

4. What is reagenic antibody ?

5. Write the role of Histamine in anaphylaxsis.

6. List out the HLA antigen role in graft rejection.

7. How do tumor cells escape from immune response ?

8. Define Allograft.

9. What is serum sickness ?

10. What is Di George syndrome ?

Section B (5 × 5 = 25)

Answer **all** questions.

11. (a) Differentiate the Active and Passive immunity.

(*Or*)

(b) Write a short note on structure and function of Lymphnode.

12. (a) Give an account on properties of Antigens.

(*Or*)

(b) Write a note on principle and uses of ELISA.

13. (a) Describe the mechanism of alternative complement pathway.

(*Or*)

(b) Give an account on MHC types and functions.

14. (a) Describe the mechanism and mediators of Anaphylaxis.

(*Or*)

(b) What is immune response ? Explain the various immunological cells in cell mediated immunity.

15. (a) Narrate the role of Lymphokines in cellular immune response.

(*Or*)

(b) Write a note on autoimmune diseases.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the maturation and differentiation of T–Lymphocytes.

17. Briefly discuss the production of Monoclonal antibody.

18. Discuss in detail about antigen processing and presentation.

19. Describe the transplantation types with suitable examples.

20. Explain the following vaccines :

(a) Toxoid vaccine.

(b) DNA vaccine.

(c) Edible vaccine.

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AF-3968 **MMI3E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Microbiology

Elective III—NANO SCIENCE AND TECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Bionanomachines.

2. Protein-nanomaterial.

3. X-ray diffraction.

4. CAMD.

5. Biomolecular structure.

6. Colorimetric vesicles.

7. Nanorobots.

8. Nanofabrication.

9. Computer chips.

10. Nanozurveillance.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Explain the special structural role of polysaccharides.

(*Or*)

(b) Give a brief account on the present status of bionanotechnology.

12. (a) Write short notes on X-ray Crystallography.

(*Or*)

(b) Describe the role of NMR spectroscopy in nano - science.

13. (a) Give an account on natural bionanotechnology.

(*Or*)

(b) Explain the flexibility of protein structure and function.

14. (a) Write short notes on functional self application.

(*Or*)

(b) Give an account on Machine-phase bionano technology.

15. (a) Describe the Nanotubes synthesis.

(*Or*)

(b) Give a brief account on nanoscale assembly and applications.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Give an account on bionanomaterials and their roles.

17. Write in detail about computer modelling for simulation of structure and function of nanoscale materials.

18. Describe the prediction of biomolecular structures.

19. Write notes on : (a) Chemical transformation.

    (b) Biomolecular sensing.

20. Explain the problems and the methods of solving them in bionanotechnology.

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AF-3969 **MMI4C1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Fourth Semester

Microbiology

RECOMBINANT DNA TECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions

1. What is the fucntion of T4 DNA polymerase ?

2. Polynucleotide kinase.

3. pET vector.

4. Give some examples for gene fusion vectors

5. Coomb's test.

6. Define zoo blot.

7. AP-PCR.

8. Dendrogram.

9. Microfluidic Sanger sequencing.

10. PHRAP.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Diagrammatically illustrate the fucntion of klenow fragment.

(*Or*)

(b) Write short notes on restriction modification systems.

12. (a) Discuss the fucntion of pBluescript vector.

(*Or*)

(b) What is meant by two- hybrid screening ?

13. (a) Give an account on colony hybridization method.

(*Or*)

(b) Describe on the mechanism of shotgun cloning.

14. (a) List out the general rules in primer designing.

(*Or*)

(b) Discuss on applications of PCR in forensic science.

15. (a) How solid sequencing technology differs from other sequencing methods ?

(*Or*)

(b) Briefly describe about chemical sequencing method.

Part C (3 × 10 = 30)

Answer any **three** of the following questions.

16. Illustrate on the fucntion and types of DNA dependent RNA polymerase in pro and eukaryotes.

17. Describe on commercially available sequencing vectors with suitable diagrams.

18. Elaborate on the methods on screening recombinants.

19. Discuss in detail the strategies involved in PCR primer design and list some of the free and commercial softwares for primer desining.

20. Write an essay on the contig alignment softwares used in sequencers.

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AF-3970 **MMI4C2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Fourth Semester

Microbiology

FOOD AND DAIRY MICROBIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. List the molds present in the food.

2. Name the thermophilic bacteria in food.

3. What are the foods can preserve under anaerobic conditions ?

4. How can you apply the natural agents as preservatives ?

5. Name the fish pathogenic bacteria.

6. Define pasteurization.

7. Define Salmonellosis.

8. Comment of FSA.

9. What is Kefir ?

10. Report on genetically modified foods.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Describe the importance of lactic acid bacteria in food industry.

(*Or*)

(b) Give the general characteristics and importance of molds in food industry.

12. (a) Discuss the preservation methods of food by heat.

(*Or*)

(b) Give an account on applications of canning in food industry.

13. (a) Describe the spoilage of poultry products and its characterization.

(*Or*)

(b) Give an account on bacterial spoilage in canned foods.

14. (a) Discuss about Hazard Analysis and Critical Control Points as a method to ensure Microbiological safety of foods.

(*Or*)

(b) What is enterotoxin ? Give the toxicity of bacterial enterotoxins.

15. (a) Explain the steps involved in cheese production.

(*Or*)

(b) Detailed account of microbial indicators of food quality and safety.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe in detail the factors influencing microbial growth in food.

17. Explain the principles of food preservation. Discuss preservation of food by chemical methods.

18. Write an essay on the contamination and spoilage of milk and milk products.

19. Elaborate - food borne diseases.

20. Discuss in detail about milk based fermented products.

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AF-3971 **MMI4E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Microbiology

Elective : IPR, BIOSAFETY AND BIOETHICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. International Trade Organisation.

2. IBSC.

3. Copyfraud.

4. Plagiarism.

5. Superbug.

6. Trademark.

7. FDA.

8. RABS.

9. Transgenic Organism.

10. ES cell lines.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write the principles and functions of WTO.

(*Or*)

(b) Describe the role of RCGM in DBT.

12. (a) Discuss the trade related aspects of IPR.

(*Or*)

(b) Explain the enforcement and limitation of patents.

13. (a) What are the rules for patent licensing ?

(*Or*)

(b) Give a brief account on Basmati case study.

14. (a) Give an introduction to the biosafety guidelines for rDNA research.

(*Or*)

(b) Write the history and rationale of biosafety level.

15. (a) Describe the purpose, scope and principles of bioethics.

(*Or*)

(b) Discuss how bioethics developed as a discipline.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain the establishment and functions of GATT.

17. Write in detail about the different types of IPR.

18. Discuss in detail about the rules governing patents.

19. Give a detailed account on GLP.

20. Bring out the issues related to GM microorganisms and their environmental release.

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AF-3972 **MMI5E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Microbiology

Elective : BIOSTATISTICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Random sample.

2. Bar diagram.

3. Cartogram.

4. Range.

5. Frequency curve.

6. Kurtosis.

7. Probability.

8. Standard error.

9. Regression.

10. t-test.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Explain the measures of Central tendency.

(*Or*)

(b) How do you find the Grouped mean and Grouped variance ?

12. (a) Describe a table and the specific uses.

(*Or*)

(b) Write short notes on graph and its types.

13. (a) Give an account on confidence interval of a mean and between means.

(*Or*)

(b) Explain standard deviation and its applications in biostatistics.

14. (a) What is a 'P' value ? What are the misunderstandings about 'P' value ?

(*Or*)

(b) How Baye's theorem is used to calculate the carrier probabilities in genetics ?

15. (a) Explain Simple correlation and uses of Scatter diagram.

(*Or*)

(b) Describe the test of significance and uses.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. How will you find the mean of grouped data and variance ?

17. Give an account on histograms and frequency polygon with their uses.

18. Explain the confidence interval for the difference or ratio of two properties.

19. Describe the Hardy-Weinberg equation and its applications.

20. Write in detail the statistical hypothesis testing and importance.

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